1 Overview: introduction to Earth & planetary climates

Hi everyone! Welcome to C5: Physics of Atmospheres & Oceans. Our agenda for our first class meeting today is the following:

1. Review the course format and structure (10 minutes)
2. Do a warm-up activity on fundamental planetary energy balance (15 minutes)
3. Broad-brush review of Earth's climate and other planetary climates (30 minutes)

I will make a notes handout for each class and post it on Canvas beforehand alongside the slides, listing the plans for the day as well as providing some further information and optional readings. These notes will not necessarily be comprehensive, and will instead typically be a summary of what is presented in class (occasionally it'll be the opposite, and they'll contain more information than we end up covering). I welcome feedback on these notes, as I consider them a living document that I will iterate on each year.

Before each lecture, I encourage you to send me an email with one (or more) questions that you have on the material either from the last lecture or the notes/reading from the upcoming lecture on Canvas. I will use these questions to better prepare each lecture, and I will often answer these questions directly during class when they are instructive for everyone.

The optional reading for Day 1: Pierrehumbert Principles of Planetary Climate (PPC), Chapter 1 (especially Sections 1.1-1.6). Please note that I am *not* following this textbook closely (or in the same order), but I think that it is the best possible reference for the material we will be covering.

1.1 Course Overview

The primary objective of this course is: *To learn how to use the basic building-blocks of physics to understand phenomena related to climate and weather on Earth and other planets, including the growing catalogue of exoplanets.* As a result, the focus of the course is to understand the fundamental physics that governs planetary climate. The course is largely Earth-centric, but these fundamental principles will be taught in a manner that can be generalised to apply to all other planets.

These course notes will cover the first sixteen lectures of C5. They are organised as follows:

1. Synopsis of planetary climates, big questions in planetary climate (1 lecture)
2. Thermodynamics of planetary atmospheres, both dry and moist ($\approx$ 4 lectures)
3. Radiative transfer of planetary atmospheres ($\approx$ 6 lectures)
4. Climate sensitivity ($\approx$ 2 lectures)
5. Clouds and their feedback on climate ($\approx$ 3 lectures)

Each section of these lecture notes corresponds to an individual lecture. Please note that these lecture notes do not cover the other material covered in the next half of the course, which includes geophysical fluid dynamics and climate dynamics, both covered in Hilary term.

1.1.1 Recommended textbooks

The course material for the thermodynamics and radiative transfer segment of C5 will largely follow the notes presented here rather than any individual textbook. That said, the primary recommended textbook for the thermodynamics and radiative transfer modules of C5 is Prof. Ray **Pierrehumbert's Principles of Planetary Climate (Cambridge University Press, 2010)**. Other recommended textbooks include:

1. Goody & Young, Atmospheric Radiation: Theoretical Basis (2nd edition), Oxford University Press, 1989 (highly recommended).

2. Liou, An Introduction to Atmospheric Radiation (2nd edition), Academic Press, 2002.

3. Peixoto & Oort, Physics of Climate, American Institute of Physics, 1992 (better for climate dynamics, but does include some fundamental thermodynamics and radiation).

4. Seinfeld & Pandis, Atmospheric Chemistry and Physics: From Air Pollution to Climate Change (2nd edition), Wiley, 2006 (more focused on chemistry and aerosols, but also includes relevant fundamental thermodynamics and radiation).

5. Lohmann, Lüönd, Mahrt, An Introduction to Clouds, Cambridge University Press, 2016 (comprehensive text on moisture and clouds)

6. Sánchez-Lavega, An Introduction to Planetary Atmospheres, CRC Press, 2011 (good introduction to atmospheres throughout the Solar System).

7. Seager, Exoplanet Atmospheres, Princeton University Press, 2010 (comprehensive introduction to exoplanet atmospheres).

8. Heng, Exoplanetary Atmospheres: Theoretical Concepts and Foundations, Princeton University Press, 2017 (exoplanets text with more detailed descriptions of radiative transfer, scattering, and chemistry).

1.2 Energy balance activity

As a warm-up after the long summer break, please get into groups of two (or three) to work on the following questions.

1. *Why doesn't the Earth melt?* Earth receives a stellar flux of approximately $F_\star = 1360$ W m^{-2} from our Sun over a disk of area πa^2, where $a = 6371$ km is the radius of Earth. Assume Earth is an isothermal perfect sphere with a density of 5500 kg m^{-3} and a specific heat capacity of 840 J kg^{-1} K^{-1}. Write down a simple differential equation that governs the change in temperature of Earth with time, assuming that heat is mixed uniformly throughout the planet. How long would it take Earth to reach a temperature of 2500 K, approximately the melting point (liquidus) of rock?

2. *What is Earth's radiatively stable temperature?* Radiation to space is the fundamental stabilising feedback that keeps Earth from melting and more generally keeps planetary temperatures at finite values. In reality, the Earth (and other planets) is in a top-of-atmosphere radiative balance between outgoing longwave radiation (OLR, i.e., thermal emission) and received stellar flux:

$$\pi a^2 F_\star (1 - \alpha) = 4\pi a^2 \sigma T_{\rm eff}^4, \tag{1.1}$$

where $\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4} is the Stefan-Boltzmann constant, $T_{\rm eff}$ is the planetary emitting temperature, and α is the planetary albedo. Given the values from Question 1 above, plug in and solve for the radiative equilibrium temperature of Earth for both cases with zero albedo ($\alpha = 0$) and $\alpha = 0.3$.

1.3 Big questions in planetary climate

There are myriad unsolved questions about planetary climate. Here are a few of the largest-scale open questions[1], to give you a sense of the scale of the problems we consider here (roughly in order of appearance in C5):

1. How do planetary atmospheres form, and what are they made of? Are atmospheres primarily inherited from the Solar nebula, built up from volatiles outgassed from the planetary interior, and/or due to the addition of volatiles from elsewhere?

2. What determines the thermal structure of planets, including the surface temperature, the vertical dependence of temperature on height, and the spatial variation of temperature with latitude and longitude?

3. How sensitive are climates to perturbations? What are the conditions under which climates can be unstable?

4. What controls the behaviour of clouds, and how do clouds affect planetary climate?

5. What are the possible atmospheric circulation regimes that can exist on a given planet, and what controls the boundaries between these regimes?

6. What are the mechanisms driving climatic structures (e.g., jet streams, Hadley cells) and weather (e.g., thunderstorms, tropical cyclones) in planetary atmospheres and oceans?

7. What can we learn about planetary climate from remote observations, including orbiting satellites (for Earth and Solar System planets) as well as telescopes (for exoplanets)?

1.4 Planetary climates in our Solar System and beyond

Below is a very brief survey of the climates of planets and moons in our Solar System, as well as a discussion of the potential climates of exoplanets.

[1] Credit: Adam Showman

1.4.1 Terrestrial bodies

There are four terrestrial bodies in our Solar System with a thick atmosphere: our own Earth, Venus, Mars, and Saturn's moon Titan. We briefly survey each of these climates as follows.

Earth

Earth is the largest terrestrial (rocky) planet in the Solar System. Some relevant quantities to note:

1. Earth mass $M_\oplus = 5.97 \times 10^{24}$ kg
2. Earth radius $R_\oplus = 6371$ km (Volumetric mean radius)
3. Rotation period, Solar (with respect to the Sun): $P_{\text{rot},\odot} = 1$ day $= 86400$ sec
   Rotation period, sidereal (with respect to background stars): $P_{\text{rot},\star} = 86164.1$ sec $=$ 23 hr 56 min 4.1 sec
4. Obliquity $\epsilon_\oplus = 23.5^\circ$
5. Average surface pressure $P_\oplus = 1$ bar $= 10^5$ Pa,
   average surface temperature $T_\oplus = 288.25$ K (2024)
6. Atmospheric composition: 78% N_2, 21% O_2, 0.9% Ar, 0.0428% CO_2 (August 2026[2]), 0-1% H_2O (depending on location), 1939 ppb CH_4 (May 2026[3]).

Earth is the only known planet with life that is affecting its present-day atmospheric composition, including the oxygen-rich nature of the atmosphere (which began $\sim$ 2.5 Ga at the great oxygenation event, GOE, and underwent another large uptick associated with the Cambrian Explosion at $\sim$ 550 Ma), as well as the increasing abundances of carbon dioxide and methane due to anthropogenic (human-induced) emissions.

The Earth has a temperature structure with many distinct layers, shown in Figure 1.1. This temperature structure has led to several standard definitions for atmospheric layers, which starting from the bottom are termed the *troposhere, stratosphere, mesosphere, and thermosphere*. Temperature decreases with height in the troposphere and mesosphere, and increases with height in the stratosphere and thermosphere. Curiously, the tropopause (the boundary between the troposphere and stratosphere) occurs at a pressure of $\sim$ 0.1 bar on Earth and all Solar System planets with thick atmospheres (Robinson & Catling, 2014). Planetary scientists often also term the layer above the thermosphere the "exosphere," which is the layer where the atmosphere of a planet thins and begins to blend into space.

Figure 1.1 also shows the characteristic zonal-mean (east-west average) temperature distribution as a function of pressure and altitude. Notably, Earth has a large near-surface equator-to-pole temperature contrast due to the equator-to-pole contrast in received irradiation. This near-surface temperature contrast is the key driver of its tropospheric atmospheric dynamics, which in turn mutes the equator-to-pole temperature contrast relative to its radiative equilibrium value – this will be covered in detail in the second half of this course. In

[2] `https://science.nasa.gov/earth/explore/earth-indicators/carbon-dioxide/`
[3] `https://science.nasa.gov/earth/explore/earth-indicators/methane/`

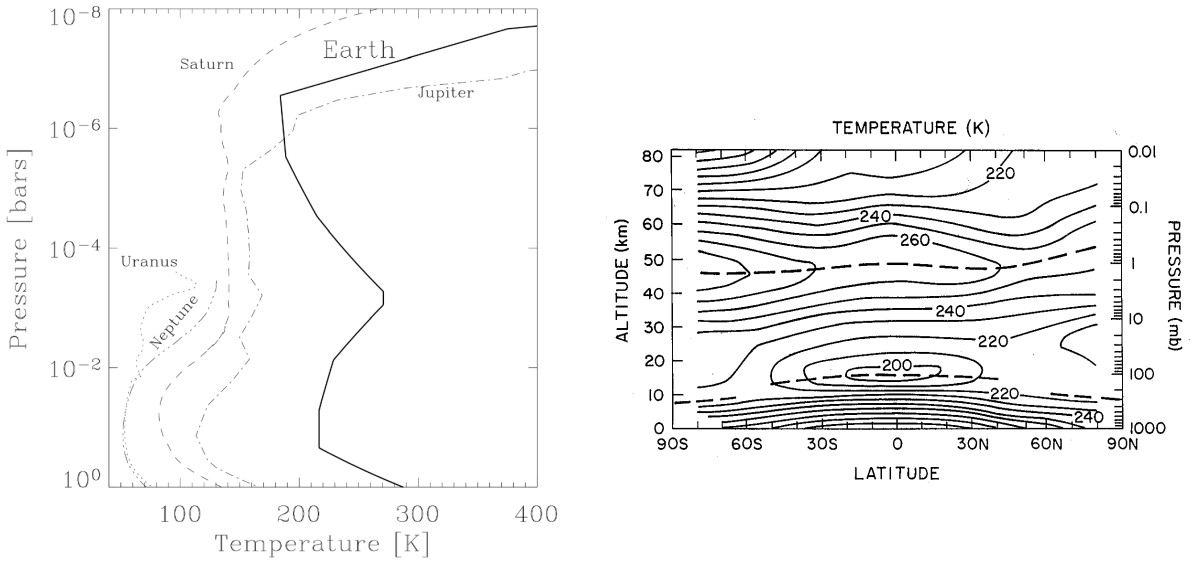

Figure 1.1: Left: Temperature-pressure profile of Earth compared to other Solar System objects with similar or cooler atmospheric temperatures. Right: Earth's zonal-mean thermal structure as a function of latitude and altitude/pressure (from Salby, 1996)

addition, note that Earth's troposphere is not uniformly thick – instead, the troposphere is ~ 15 km thick at the equator and ~ 8 km thick near the poles.

Figure 1.2 shows an image of the Earth taken by the DSCOVR mission from Lagrange point L1, 1.5×10^6 km from Earth. Earth is of course most notable for the "blue marble"

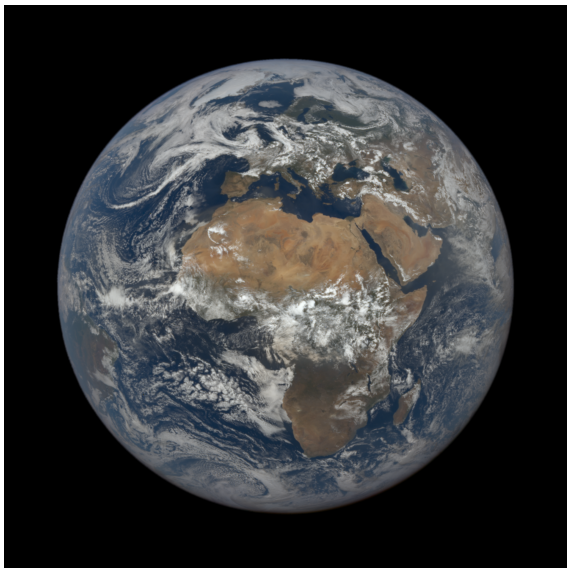

Figure 1.2: Earth from the DSCOVR mission (image taken by the EPIC instrument). Image taken from `https://epic.gsfc.nasa.gov`. One can clearly see the cloudy moist African tropics southward of the clear and dry Saharan sub-tropics.

appearance of its oceans from afar, but let's dig into this disk-resolved image in more detail. First, we can see that the tropics (especially over continental Africa) are cloudy relative to the surrounding regions (and a bit greener on the surface if you look closely). This implies a greater local amount of moisture relative to the surrounding regions (i.e., the sub-tropical Sahara and South Africa). The reason for this is due to the Hadley circulation, in which warm moist air rises near the equator, rains out, and dries as it moves poleward, sinking and

adiabatically warming at latitudes of $\sim 20 - 30^\circ$. The reason the Hadley cells do not extend further is because of Earth's relatively rapid rotation, as the Coriolis force prevents a parcel of gas being launched poleward from the equator to reach the pole. Instead, the Coriolis force deflects the parcel eastward (in either hemisphere), forming eastward sub-tropical jet streams in each hemisphere. In tandem, eddies (smaller-scale cyclones and anti-cyclones) form in the mid-latitudes, driven by the temperature contrast from the edge of the Hadley cell to the pole. These eddies and waves (especially Rossby waves) in turn generate a momentum flux that leads toward an eastward jet in the mid-latitudes. Earth just happens to have the correct rotation rate for these two jets (sub-tropical and eddy-driven) to merge (Kaspi & Showman, 2015), leading to Earth having one eastward mid-latitude jet in each hemisphere (see Figure 1.3).

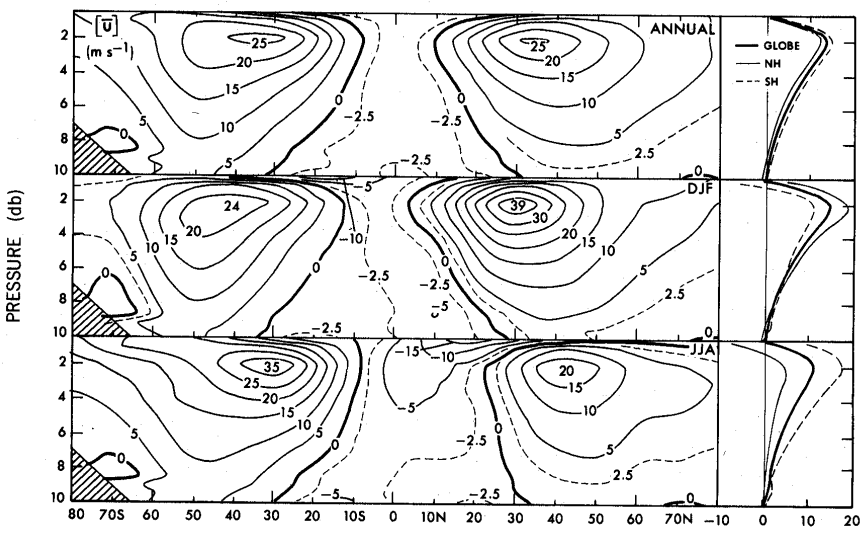

Figure 1.3: Zonal-mean zonal winds (east-west mean of the east-west winds) in Earth's atmosphere, on an annual average (top) and averaged over December-February (middle) and June-August (bottom). Figure adapted from Peixoto & Oort (1992).

Figure 1.4 shows the typical atmospheric features of an Earth-like exoplanet – note that Earth does not display all of these features, but planets like Earth with different planetary parameters (e.g., rotation rate, planetary radius, etc.) may. Specifically, the schematic displays a planet that is somewhat more slowly rotating than Earth, as it exhibits equatorial superrotation (winds at the equator are sufficiently fast that the atmosphere has greater eastward angular momentum than the solid-body rotation of the planet), similar to that in the atmospheres of the slower-rotating Venus and Titan (see the following sections for more). The Hadley cell is also slightly wider (extends to higher latitudes) than that of Earth itself, further indicative of somewhat slower rotation (Held & Hou, 1980). Finally, the subtropical and eddy-driven jets are distinct rather than merged into one, as the eddy-driven jet is preferentially formed by Rossby wave stirring at higher latitudes (Kaspi & Showman, 2015). Note also the distinction made in the diagram between the dynamical "tropics" and "extratropics." The dividing line between the tropics and extratropics typically corresponds to when the local Rossby number

$$\mathrm{Ro} \equiv \frac{U}{fL} = 1, \tag{1.2}$$

where the Rossby number represents the ratio of the advective to the Coriolis term in the momentum equation, with $f = 2\Omega \sin\phi$ the Coriolis parameter as a function of rotation rate

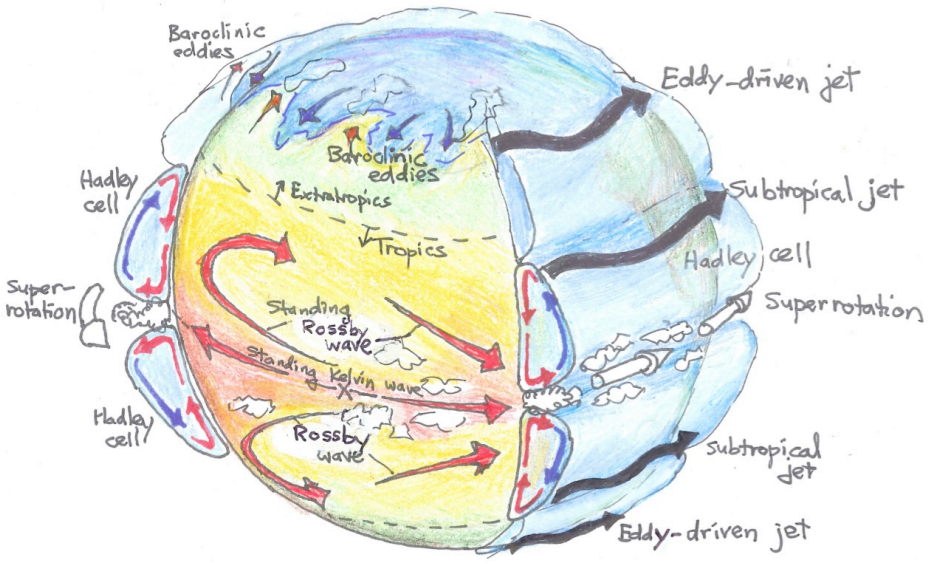

Figure 1.4: Schematic showing the key features of the atmospheric circulation of an Earth-like planet. Figure adapted from Showman et al. (2013).

Ω and latitude ϕ. Regions are dynamically "tropical" if Ro > 1 and the Coriolis force is weak relative to advection, and "extratropical" if Ro < 1 and the Coriolis force is strong relative to advection. Dynamically tropical regions are typically near the limit with weak temperature gradients (WTG) (Bretherton & Sobel, 2003), while the extratropics have large temperature contrasts that provide the energy to drive baroclinic eddies which in turn transport heat from the edge of the Hadley cell toward the pole.

Other terrestrial bodies: Venus, Mars, and Titan

Our Solar System has three other terrestrial bodies beyond Earth that have thick atmospheres – two are planets (Venus and Mars) and one is a moon of Saturn (Titan). Figure 1.5 shows the characteristic atmospheric properties of all three of these bodies including Earth. Venus is by far the hottest of the four – this is not simply because it is closest to the Sun (note that because of its high albedo, Venus has a *lower* equilibrium temperature than Earth). Instead, it is due to Venus having experienced a runaway greenhouse and extreme water loss (Ingersoll, 1969), as evidenced by its high atmospheric D/H ratio. Venus' atmosphere is much thicker than that of the other three objects, with a surface pressure of 92 bars and an atmosphere composed almost entirely of CO_2 (see Figure 1.6). Mars also has a CO_2-rich atmosphere, but it is much cooler than Venus due to its thin atmosphere with a surface pressure of 6 mbar. Mars has the most Earth-like meteorology of any planet in the Solar System, largely due to its similar rotation period of 24.6 hours and current obliquity of 25°. Titan has the closest atmospheric thickness and composition to Earth of any object in the Solar System, with a surface pressure of 1.47 bars and atmosphere comprised largely of N_2. However, Titan's atmosphere is much colder than Earth's due to its large mean separation of 9.5 au from the Sun, implying an equilibrium temperature sufficiently cold that its crust is comprised of water ice.

Each of these three terrestrial bodies have rich atmospheric phenomena, with entire subfields devoted to their study. For instance, Mars undergoes strong seasonal changes in its atmospheric thickness due to the deposition (in winter) and sublimation (in spring) of CO_2 ice at its polar ice caps (see Figure 1.7). In addition, Mars regularly undergoes global-scale dust

Table 4.2 Basic atmospheric parameters for Venus, Earth, Mars and Titan.

Parameter	Venus	Earth	Mars	Titan	Reference
Mean heliocentric distance (AU)	0.723	1.000	1.524	9.543	1
Geometric albedo $A_{0,v}$	0.84	0.367	0.15	0.21	1, 3, 4
Bond albedo	0.75	0.306	0.25	0.20	1, 2, 3, 4, 5
Surface temperature (K)	737	288	215	93.7	1, 2, 3
Equilibrium temperature (K)	232	255	210	85	Calc.[a]
Exobase[b] temperature (K)	270–320	800–1250	200–300	149	2, 6, 7
Surface pressure (bar)	92	1.013	0.00636	1.47	1, 2, 3
Scale height at surface (km)	16	8.5	11	20	Calc.[c]

[a] Calculated with eq. (3.17) and $\epsilon = 1$. The globally and wavelength averaged emissivity for Earth is 0.96–0.98.

[b] A range of values is given for Venus, Earth and Mars, appropriate for a range of solar (low to high) activities.

[c] Calculated with eq. (4.2).

1: Yoder (1995), and http://nssdc.gsfc.nasa.gov/planetary/. 2: Chamberlain and Hunten (1987). 3: Fulchignoni *et al.* (2005). 4: Moroz (1983). 5: Hunten *et al.* (1984). 6: Waite *et al.* (2005). 7: Forbes *et al.* (2008).

Figure 1.5: Atmospheric physical properties of Venus, Earth, Mars, and Titan. Table taken from de Pater & Lissauer (2001).

storms every few Martian years (and many more smaller-scale dust storms more regularly), as the planet is very dry given that it is too cold to presently have stable liquid water on its surface. Radiative feedbacks allow dust to interact with dynamics, as the dusty air absorbs radiation over a broader range of wavelengths than clear air, leading to the dusty air being heated and imposing large temperature and pressure gradients that lift the dust and drive more heating. This positive feedback then can lead to rapid growth of planetary-scale dust storms.

Titan is shrouded in a layer of haze, preventing observation of the surface at visible wavelengths (see Figure 1.8). However, spacecraft observations have enabled more detailed study of the atmosphere of Titan, especially with the Cassini Saturn orbiter and Huygens entry probe. Most notably, Cassini/Huygens discovered that methane/ethane forms liquid lakes on Titan's surface, and more generally that these organics play the same role of water on Earth as a condensible species that is also stable as a liquid on the surface. Importantly, the chemical destruction timescale of methane by photolysis on Titan is predicted to be on the order of ~ 10 Myr (which in turn is expected to form the photochemical haze layer seen in visible-light images). This requires an unknown source replenishing this methane into the atmosphere, possibly deep-seated aquifers.

Though Venus is often thought of as Earth's twin, out of all the terrestrial bodies in the Solar System with thick atmospheres it has the atmosphere most dissimilar to that of Earth. Venus' atmosphere is covered in a thick layer of sulfuric acid clouds (see Figure 1.9), preventing study of the surface at visible light (like Titan). Venus further lies in a very different dynamical regime than Earth, with a thick atmosphere and a very slow (sidereal) rotation period of 243 Earth days (with a 117 Earth day Solar day). This slow rotation and thick atmosphere leads to very weak temperature gradients of only a few Kelvin in the deep atmosphere of Venus. In addition, Venus has very strong eastward (superrotating) winds at the cloud level, as shown by the "V feature" in the left-hand panel of Figure 1.9. Venus' atmospheric structure and circulation provides a bridge between the studies of terrestrial

Table 4.4 Atmospheric composition of Earth, Venus, Mars and Titan.[a]

Constituent	Earth[b]	Venus	Mars	Titan[c]	References
N_2	0.7808	0.035	0.027	~0.95	1, 2, 3, 4
O_2	0.2095	0–20 ppm	0.0013		1, 2, 5, 6
CO_2	345 ppm	0.965	0.953	10 ppb	1, 2, 4
CH_4	2 ppm		10–250 ppb	0.014–0.049[d]	2, 3, 4, 7, 8, 9
H_2O	<0.03[f]	30 ppm[e]	<100 ppm[f]	0.4 ppb	1, 10, 11
H_2O_2	1 ppb		18 ppb		2, 12
Ar	0.009	70 ppm	0.016	28 ppm	2, 3, 4, 8
CO	0.2 ppm	20 ppm[e]	700 ppm	45 ppm	2, 11, 13
O_3	~10 ppm[g]		0.01 ppm		1, 2
C_2H_2	8.7 ppb			189 ppm	14, 15
C_2H_6	13.6 ppb			121 ppm	14, 15
C_3H_8	18.7 ppb			0.6 ppm	14, 15, 16
C_2H_4	11.2 ppb			40 ppm	14, 15
C_3H_4				3.9 ppm	14
NO	<0.5 ppb		3 ppm		2
N_2O	0.35 ppm				1
SO_2	<2 ppb[f]	100 ppm			11, 15
H_2SO_4		1–2.5 ppm			17
H_2	0.5 ppm		10 ppm	0.004	2, 14
HCl		0.1 ppm[h]			11
HF		2 ppb[h]			11
COS		4 ppm			11
He	5 ppm	12 ppm			2
Ne	18 ppm	7 ppm	2.5 ppm	<0.01	2
Kr	1 ppm	0.2 ppm	0.3 ppm	<0.1 ppb	2, 5, 8
Xe	0.09 ppm	<0.1 ppm	0.08 ppm	<0.1 ppb	2, 8

Figure 1.6: Atmospheric composition of Venus, Earth, Mars, and Titan. Table taken from de Pater & Lissauer (2001).

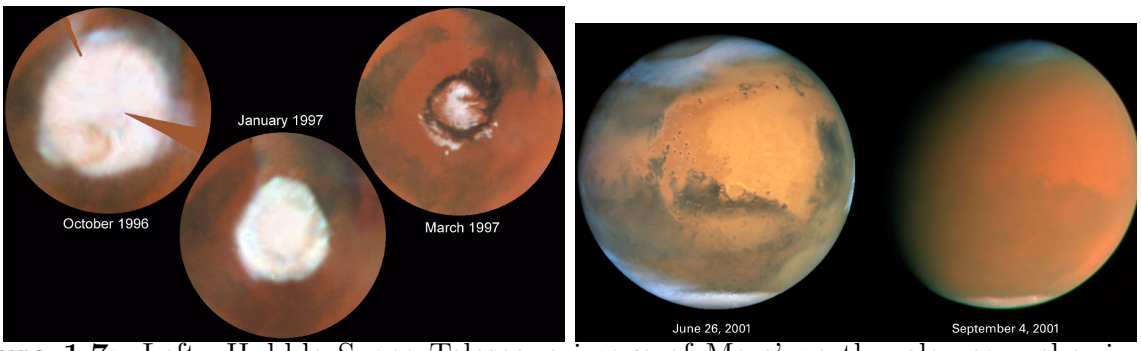

Figure 1.7: Left: Hubble Space Telescope image of Mars' north polar cap, showing its maximum and minimum extent as the CO_2 polar cap sublimates when winter transitions to spring. Right: The onset of a global dust storm on Mars in 2001, also observed with Hubble.

bodies with relatively thin atmospheres and more massive planets with thick atmospheres.

1.4.2 Gas and ice giants

Our Solar System has four planets beyond the asteroid belt which are completely fluid and have no observable solid surface: Jupiter ($11.2R_\oplus$), Saturn ($9.4R_\oplus$), Uranus ($4.0R_\oplus$),

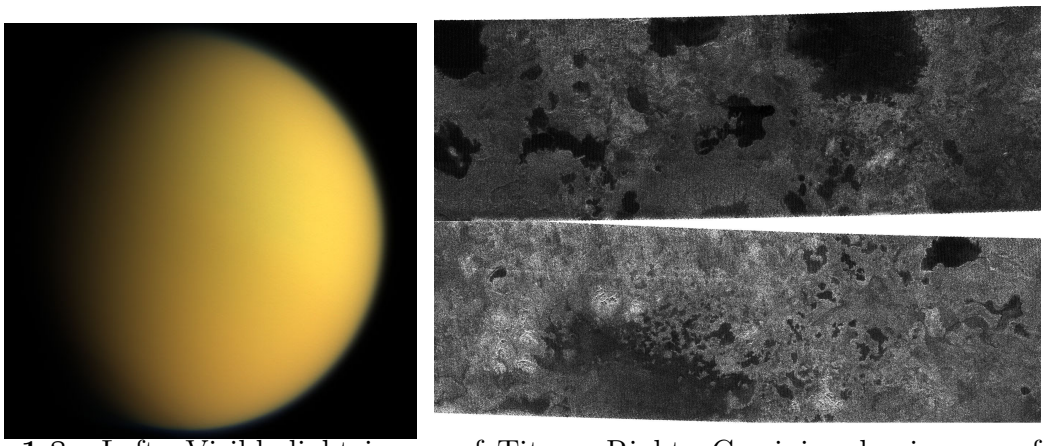

Figure 1.8: Left: Visible-light image of Titan. Right: Cassini radar images of Titan's surface, showing dark features that are likely lakes of ethane/methane on the surface.

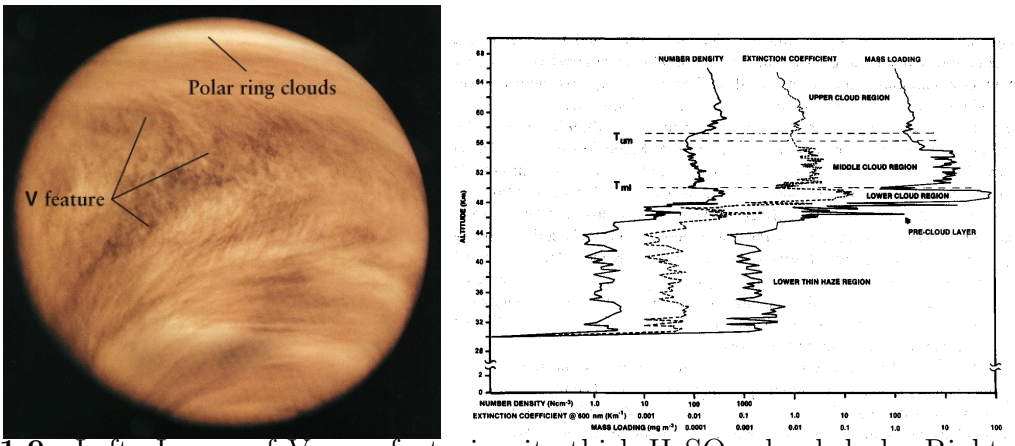

Figure 1.9: Left: Image of Venus, featuring its thick H_2SO_4 cloud deck. Right: Vertical structure of Venus' cloud decks, adapted from Esposito et al. (1983).

and Neptune ($3.9R_\oplus$). Figure 1.10 shows the characteristic atmospheric properties of the Solar System gaseous planets. The atmospheres of all these planets are dominated by H_2 and He inherited from the Solar nebula. However, each of these atmospheres are differentially enhanced in heavy elements with respect to our Sun.

Figure 1.11 shows elemental abundances of various species on Jupiter, Saturn, Uranus, and Neptune. One can see that Jupiter has the lowest abundance ratios of $\sim 3\times$ Solar for all species, the species observed for Saturn are $\sim 10\times$ Solar, and carbon on Uranus and Neptune (inferred from CH_4) is $\sim 100\times$ Solar. Note that these abundance ratios are often referred to as the "metallicity" with respect to Solar. Thus, though the atmospheres of all four planets are largely H_2, the bulk composition of Uranus and Neptune is distinct from that of Jupiter and Saturn. The enrichment of astrophysical "ices" (e.g., carbon-bearing, nitrogen-bearing, and oxygen-bearing species like CH_4, NH_3, and H_2O) in Uranus and Neptune (not just from the atmospheric abundances but also their densities and moments of inertia) is why they are often termed "ice giants" while the lower-metallicity Jupiter and Saturn are referred to as "gas giants."

Figure **??** in the following Chapter shows temperature-pressure profiles of the gas and

Parameter	Jupiter	Saturn	Uranus	Neptune	References
Mean heliocentric distance (AU)	5.203	9.543	19.19	30.07	1
Geometric albedo ($A_{0,v}$)	0.52	0.47	0.51	0.41	1
Geometric albedo ($A_{0,ir}$)	0.274 ± 0.013	0.242 ± 0.012	0.208 ± 0.048	0.25 ± 0.02	3
Bond albedo	0.343 ± 0.032	0.342 ± 0.030	0.290 ± 0.051	0.31 ± 0.04	3
Phase integral	1.25 ± 0.10	1.42 ± 0.10	1.40 ± 0.14	1.25 ± 0.10	3
Effective temperature (K)	124.4 ± 0.3	95.0 ± 0.4	59.1 ± 0.3	59.3 ± 0.8	2
Equilibrium temperature (K)	110	81	58	46	Calc.[b]
Temperature ($P = 1$ bar) (K)	165.0	134.8	76.4	71.5	5
Tropopause temperature (K)	111	82	53	52	5
Mesosphere temperature (K)	160–170	150	140–150	140–150	4
Exobase temperature (K)	900–1300	800	750	750	6, 7
Tropopause pressure (mbar)	140	65	110	140	5
Scale height (at 1 bar) (km)	24	47	25	23	Calc.[c]
Adiabatic lapse rate (K/km)	1.9	0.84	0.85	0.86	4
Energy balance[a]	1.67 ± 0.09	1.78 ± 0.09	1.06 ± 0.08	2.61 ± 0.28	2

[a] Ratio (energy radiated into space)/(solar energy absorbed).

[b] Calculated with eq. (3.17) and $\epsilon = 1$.

[c] Calculated with eq. (4.2).

1: Yoder (1995). 2: Hubbard *et al.* (1995). 3: Conrath *et al.* (1989b). 4: Chamberlain and Hunten (1987). 5: Lindal (1992). 6: Atreya (1986). 7: Bishop *et al.* (1995).

Figure 1.10: Atmospheric properties of Jupiter, Saturn, Uranus, and Neptune. Table taken from de Pater & Lissauer (2001).

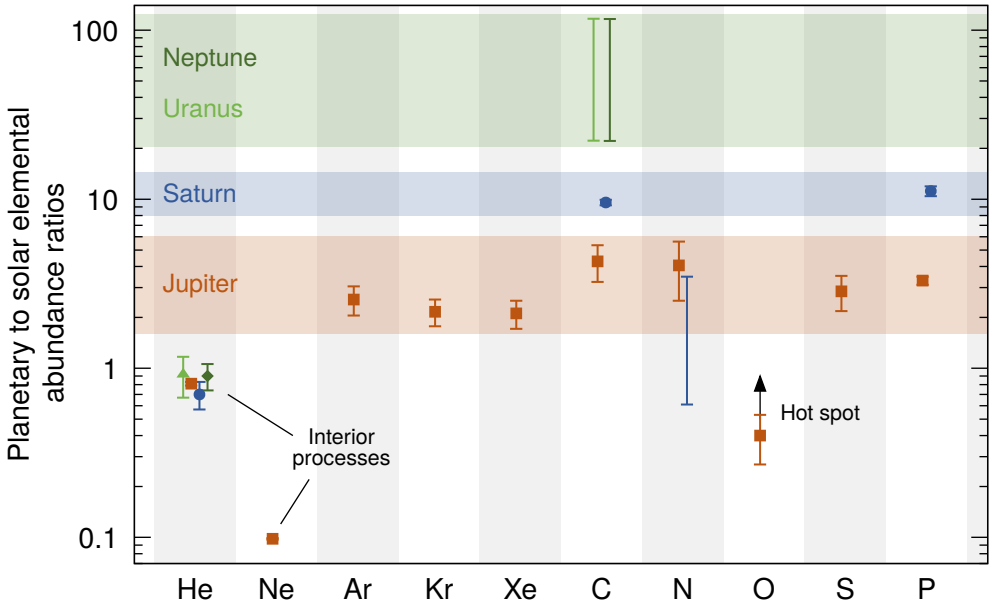

Figure 1.11: Elemental abundances of various species in the atmospheres of Jupiter, Saturn, Uranus, and Neptune. Measurements are with respect to Solar abundance of each species. Elemental abundances for Jupiter are largely from in-situ observations with Galileo, while those for Uranus and Neptune are solely through spectroscopic observations of CH_4. However, note that CH_4 condenses in the atmospheres of Uranus and Neptune, complicating this measurement and increasing its uncertainty. Figure taken from Mousis et al. (2018).

ice giants compared to the terrestrial bodies with thick atmospheres. As for the terrestrial bodies, the gas and ice giants have tropospheres at depth that transition to stratospheres at a pressure of ~ 0.1 bar. Additionally, the gas and ice giants show relatively small latitudinal temperature variations at the cloud-tops of only a few K, unlike Earth and Mars which have large thermal variations due to the equator-to-pole irradiation contrast.

The circulation of the gas and ice giants is further distinct from one another. Figure 1.12 shows the zonal-mean zonal wind inferred from cloud tracking observations of each planet[4]. Jupiter and Saturn both show equatorial superrotation, while Uranus and Neptune are sub-

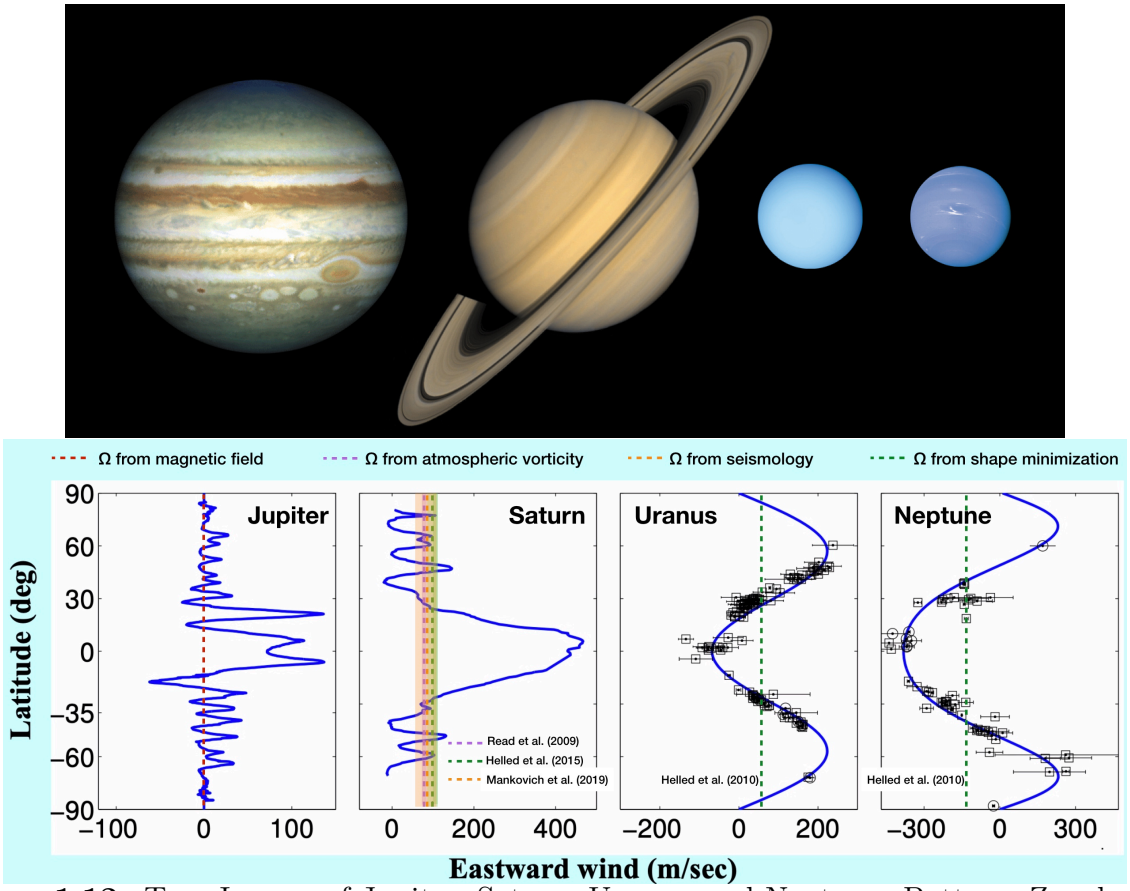

Figure 1.12: Top: Images of Jupiter, Saturn, Uranus, and Neptune. Bottom: Zonal-mean zonal winds (east west average of the east west wind) of the Solar System gas and ice giant planets from cloud tracking data. Jupiter and Saturn have superrotating equatorial regions with alternating east-west banded jets at higher latitudes, while Uranus and Neptune have sub-rotating equatorial regions and one broad eastward jet in each hemisphere. Figure taken from Guillot (2019).

rotating. Jupiter and Saturn's atmospheric circulation is banded, with alternating east-west

[4]Note that the rotation rate of each planet is typically inferred from spacecraft measurements of the time-variation of the magnetic field as it precesses, but the magnetic field of Saturn is well-aligned with its rotation axis, preventing a tight measurement of the rotation rate. Planetary scientists have used multiple methods to estimate the correct rotation rate of Saturn, leading to different resulting wind speeds which are measured with respect to the background rotation profile (corresponding to the different vertical dashed lines in the Saturn sub-panel of Figure 1.12).

jet streams. This banding on Jupiter corresponds to the alternating "belt and zone" structure due to less cloudy and darker belts and cloudier and brighter zone regions on Jupiter, where the clouds are largely comprised of NH_3, NH_4SH, and H_2O. Meanwhile, Uranus and Neptune have broad mid-to-high latitude jets, and their darker colour is largely due to the presence of CH_4 vapour with white CH_4 clouds overlain. The reason for the banded structure of these objects is due to their rapid rotation, which range from $\sim$ 10 hours for Jupiter and Saturn to $\sim$ 17 hours for Uranus.

1.4.3 Exoplanets

Our Milky Way Galaxy is teeming with planets. To date, astronomers have discovered 6,366 exoplanets[5] (see Figure 1.13). Note that the population shown in Figure 1.13 is *not*

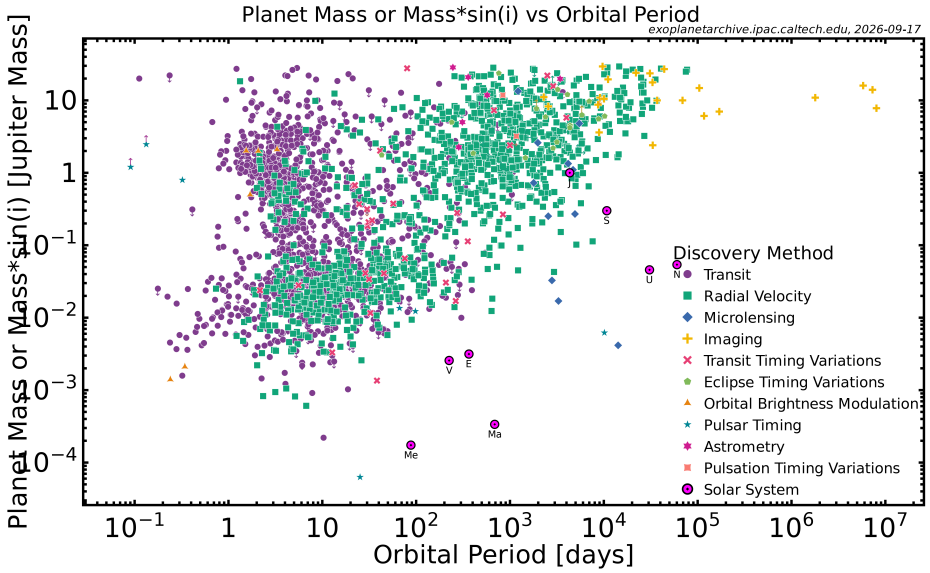

Figure 1.13: The exoplanet census as of October 2025. Note that the y-axis shows either the true planet mass, estimated planet mass, or true planet mass times the unknown sine of the inclination depending on the detection methods used to characterise a given planet (shown by the colour of each point). The Solar System planets are shown for comparison.

the intrinsic population of exoplanets. Instead, it is a biased sample due to the fact that it is easier for astronomers to detect certain types of exoplanets. For instance, the radial velocity and transit methods (the two most common methods with detections shown in the purple and teal colours in Figure 1.13) are both biased towards finding close-in and massive or large planets. In addition, the direct imaging method (taking an image of the planet by separating its light from that of the host star) is biased toward wide-separation and large planets. These three methods provide the detection of planets which are the best for follow-up characterisation – out of these, only direct imaging can (at present) find planets that are comparable to planets in our Solar System (Jupiter and Saturn in this case).

The climates of close-in planets are expected to be quite distinct from that of planets in our Solar System. This is because close-in planets with separations $\lesssim$ 0.1 au are expected

[5]value updated as of 25 Sep. 2026

to be spun-down through tidal interactions with their host star to lie in a spin-synchronous state where their rotational period is equal to their orbital period (Guillot et al., 1996). These "tide-locked" planets thus have a permanent dayside which always faces their host star and a permanent nightside looking out on the darkness of space, which greatly affects their atmospheric circulation (for a detailed review, see Pierrehumbert & Hammond, 2019).

The hottest and largest class of transiting planets are termed "hot Jupiters," which by definition are gas giant planets with equilibrium temperatures $\geqslant 1000$ K. These are the best-observed close-in planets due to their hot temperatures and large radii enabling multiple types of observations to characterise their atmospheric properties. Figure 1.14 shows a schematic of the atmospheric circulation of these planets, alongside an early observation of the hot Jupiter HD 189733b compared to the first general circulation model (GCM) simulation of a hot Jupiter conducted in 2002. Hot Jupiters have large radiative equilibrium

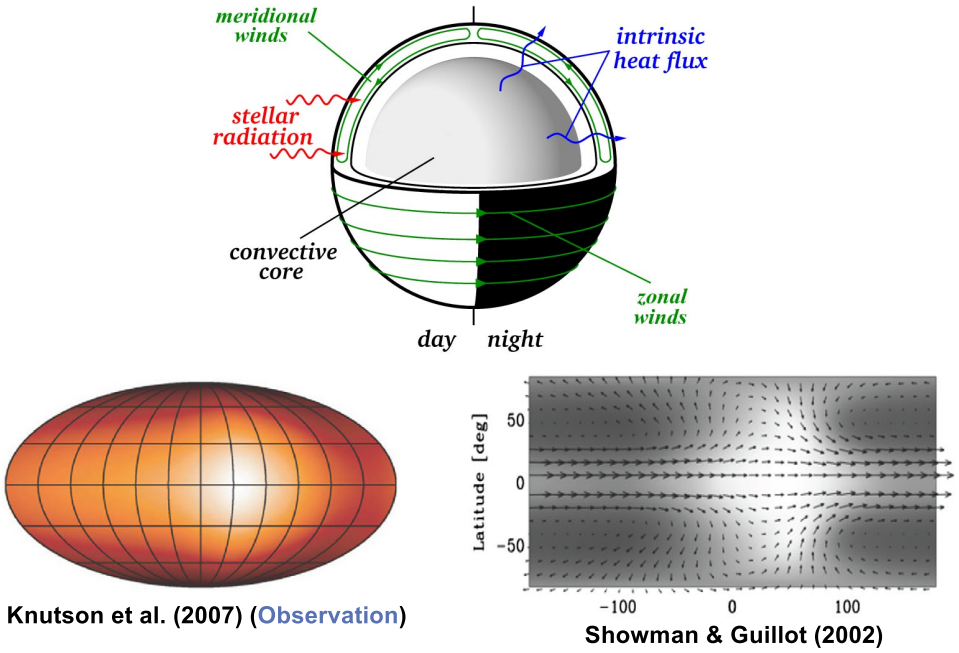

Figure 1.14: Top: Schematic of a hot Jupiter, with the host star on the left. Figure adapted from (Showman & Guillot, 2002). Bottom: Observed phase curve map of a hot Jupiter from the Spitzer Space Telescope (left) compared to a first-principles 3D general circulation model prediction of the atmospheric circulation of a hot Jupiter. Figure adapted from Showman et al. (2020).

day-night temperature contrasts, which lead to large day-to-night pressure gradients that must be balanced by flow on the order of km/s (Showman & Guillot, 2002). This flow manifests in typical simulations of hot Jupiters as a planetary-scale standing wave pattern and superrotating equatorial jet (Showman & Polvani, 2011). The equatorial jet Doppler-shifts the planetary-scale wave pattern, leading to an eastward shift of the hottest point downwind from the substellar point (Hammond & Pierrehumbert, 2018). This eastward hot spot offset has been observed for many hot Jupiters (Bell et al., 2021; Dang et al., 2025), indicative of strong atmospheric flow.

Another class of planet of interest are close-in rocky planets orbiting small, cool (M dwarf) stars. Due to the dimmer host stars, these close-in planets with orbital periods of days to tens of days can have equilibrium temperatures comparable to Earth. However, because of their close-in orbits, they are likely tide-locked to their host stars similarly to hot Jupiters. Figure 1.15 shows an example GCM simulation of a tide-locked temperate rocky planet compared to an equivalent simulation of a temperate rocky planet with a host star, orbit, and rotation period equivalent to Earth (but with no seasons). The key differences

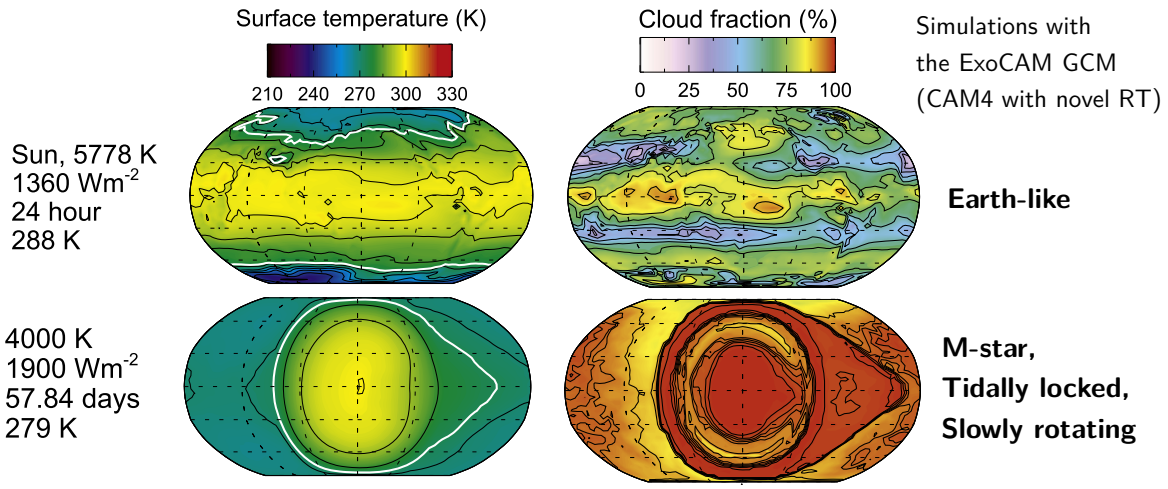

Figure 1.15: 3D general circulation model simulation of the atmosphere of an Earth-like planet with no seasons (top) compared to that of an otherwise equivalent planet with a tide-locked orbit around a smaller, cooler star with an effective temperature of only 4000 K. The left-hand panels show surface temperature, and the right-hand panels show cloud fraction. Figure adapted from (Kopparapu et al., 2017).

are that 1) in the tide-locked case, the radiative equilibrium temperature gradient driving the circulation is from day to night (sub-stellar to anti-stellar point) rather than equator to pole, 2) the rotation period of the tidally locked case is much longer, leading to a greater planetary Rossby number and weak influence of rotation on the flow. As a result, the tide-locked planet is dynamically all "tropics," with weak temperature gradients and vigorous convection throughout the dayside (rather than in an equatorial band as in the more Earth-like case shown on top). This leads to a high dayside cloud fraction in the tide-locked case, increasing the planetary albedo and cooling the global-mean temperature relative to the Earth-like case. It has been proposed that this stabilising cloud feedback on tide-locked planets could expand the habitable zones (i.e., the region around a given star where liquid water could be stable on the surface of an Earth-like planet) of M dwarf stars (Yang et al., 2013). Conversely, perhaps many rocky planets orbiting M dwarfs never underwent water condensation into surface oceans due to the bright pre-main-sequence phase of their host stars (Turbet et al., 2023).